Research Statement

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Over the past few decades, Internet use has grown precipitously around the world. Today, over 3.6 billion people use social media; we go online for music, news, television and movies, and communicating with family and friends; essential day-to-day services like shopping, banking, and even health care are increasingly delivered virtually. And the COVID-19 pandemic is only accelerating these trends. On the one hand, shifting to online services brings increases in efficiency and convenience; however, on the other hand, it has brought about an ecosystem of surveillance capitalism riddled with severe privacy threats. In its broadest strokes, my research aims to develop tools that help users protect their privacy online—without sacrificing the modern Internet's conveniences.

My research contributions to date run the gamut from low-level cryptographic innovations through the design and analysis of complex systems, all the way to heavily-optimized implementations of primitives and systems alike. My (in progress) PhD dissertation presents.

1. efficient new building blocks for secure MPC protocols,
2. scalable sigma protocols,
3. a novel system for privacy-preserving streaming with recommendations,
4. scalable sender-anonymous messaging protocols, and
5. fast implementations of all of these.

The next section elaborates on some of my key research achievements to date.

Description of the Research Carried Out in my Ph.D.

In the following I will describe the overview of the research that I carried out on the above topics. Broadly, my Ph.D. thesis involves the applications of Distributed Point Functions (DPFs) to Multi-Party Computation (MPC). Informally, DPFs are a cryptographic primitive for succinctly secret sharing a functional representation of a so-called "1-hot" vector, often just a "standard" orthonormal basis vector.

In the following I will describe two specific complex secure-systems that we designed and implemented. During the course of this work we also developed novel cryptographic primitives and wrote heavily-optimized cryptographic libraries.

Secure Systems

Pirsona Pirsona [1] is a digital content delivery system that realizes collaborative-filtering recommendations atop private information retrieval (PIR). This combination of seemingly antithetical primitives makes possible – for the first time – the construction of practically efficient e-commerce and digital media delivery systems that can provide personalized content recommendations based on their users' historical consumption patterns while simultaneously keeping said consumption patterns private. Collaborative Filtering (CF) is the most popular machine learning algorithms to provide recommendations. In our work, we develop 3- and 4-party protocols that implement gradient descent-based CF on secret shares (the parties have shares of user and item profiles). The MPC is built on top of a PIR protocol from which users fetch the items which are used by the CF to provide recommendations. Pirsona is protected against malicious users submitting malformed-DPFs via a DPF sanity check that weeds out such bad queries.

Pirsona solves a fundamentally different (in some ways harder) problem than contemporary work on MPC-based machine learning—and it does so in a surprisingly efficient way. To enable this, we developed a host of new DPF-based MPC gadgets, including

1. 2-party DPF-based DeMUX,
2. 2- and 3-party DPF-based MUXes,
3. DPF-based fixed-point vector normalization protocol,
4. DPF-based single-round integer comparison protocol,
5. 4-party sanity check to check the well-formedness of DPF, and
6. dpf++ [2] library, which implements DPFs efficiently.

The entire implementation was carried out in C++ and experiments were performed on Amazon EC2 servers. The implementation used our dpf++ library.

Sabre Sabre [3] is a Sender Anonymous Messaging (SAM) system which outperforms the state-of-the-art SAM systems, namely Riposte and Express. Sabre is more performant than prior work by a (significant) constant factor when there is no attacker, and it is *asymptotically faster* in the presence of DoS attacks. Sabre, like Riposte and Express, takes write requests from the clients in the form of DPFs. The servers need to check that the DPFs submitted are valid, to prevent a malicious server from corrupting the system.

Sabre clients use the MPC-in-the-head paradigm introduced by Ishai, Kushilevitz, Ostrovsky, and Sahai [4] to do a Zero-Knowledge Proof (ZKP), proving the validity of the DPFs they submitted; they submit the proof along with write requests (the DPFs). The MPC-in-the-head paradigm uses MPC to do ZKP. We generalize the MPC-in-the-head paradigm and present a "multi-verifier" variant of the paradigm. The main contribution of Sabre is using our novel "multi-verifier" variant of the MPC-in-the-head paradigm to do a ZKP to prove the well-formedness of the DPFs. To implement MPC-based ZKPs, we use LowMC as the block cipher.

The main contributions that were used to build Sabre are the following :

1. an efficient secure $(2+1)$-party computation for the oblivious evaluation of logarithmic-sized DPFs,
2. a new "multi-verifier" variant of MPC-in-the-head,
3. non-interactive zero-knowledge arguments of knowledge for the well-formedness of logarithmic-sized DPFs, and
4. parallel (bitsliced) implementations of fixed-key LowMC [5] and logarithmic-sized DPF evaluation (this parallel implementation allows to do 128 proofs in parallel).

Proving partial knowledge

Apart from building these systems, I have also carried out work on how DPFs can facilitate ZKPs more generally. Specifically, we will show that DPFs can be used in *efficient* ZKPs of partial knowledge. The problem setting is as follows:

- **Common Knowledge:** $C_1[= g^{x_1}], \ldots, C_n[= g^{x_n}]$.
- **Prover Knows:** $C_\ell = g^{x_\ell}$, and $\ell \in [1, \cdots, n]$

The prover's goal is to convince the verifier that they know *some* ℓ such that, $C_\ell = g^{x_\ell}$, without revealing ℓ or x_ℓ. We provide efficient ZKP for the above problem and its generalization where the prover wants to convince the verifier that they know k-out-of-n discrete logs.

Other Research

I have also demonstrated the ability to do research other areas of Computer Science during my Ph.D. During one of the semesters and summers, I worked in the Algorithms and Machine Learning Lab at Indiana University and developed an efficient distributed parallel algorithm to do Single Linkage Clustering [6]. The implementation was done in Java and Spark.

Research Plans

Short Term Goals

Computing Non-linear Functions using DPFs To develop Pirsona, we computed functions like comparison and inverse square root and comparison functions using DPFs and PIR. We believe that a similar method can be used to compute other useful non-linear functions. The goal is to study under what circumstances (properties a function has) can the PIR-based method be used to compute them. For instance, how does the entropy of a function, effect their computation using this method.

Building Other ML-based secure-systems I am also interested in designing and implementing more secure systems using cryptographic primitives like Private Information Retrieval, Multi-Party Computation, and Zero-Knowledge Proofs. In my Ph.D, I studied Collaborative Filtering (one of the most popular machine learning algorithms) in an MPC setting. I would like to securely implement other popular machine learning algorithms like for instance, ridge regression, support vector machines.

Differential Privacy based Pirsona The current implementation of Pirsona does not prevent the information leakage which is inherent due to the Collaborative Filtering algorithm. There would be a similar problem while implementing other machine learning algorithms. Another line of research I am interested in doing is using Differential Privacy to prevent the information leakage that is inherent due to the ML algorithm.

Long Term Goals

My Ph.D. focused mainly on DPFs and 1-private MPC protocols. I plan to diversify to other areas of cryptography and security, like more general MPC protocols, Zero-Knowledge Proofs, Private Information Retrieval, and Homomorphic Encryption. My main research goals are while studying these are:

a) coming up with more efficient primitives that can help build systems and protocols which provide a much stronger security to the system by anticipating and thwarting disruptive security breaches in the Internet.

b) designing, implementing, and deploying varied cryptographic systems. In particular, I am interested in working on developing systems that would provide solutions for secure elections and secure auctions.

c) studying interesting theoretical properties and problems related to DPFs, MPC, PIR, and ZKP.

References

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